Rotor-Flux Oriented Control of the Induction Machine

A Step-by-Step Derivation

1 Motivation

In indirect rotor-flux oriented control, the controller is built from a reference i_{ds}, a reference i_{qs}, and a current regulator. The torque demand sets i_{qs} directly, and the desired rotor flux level sets i_{ds}. Locating the rotating reference frame, however, requires the rotor flux angle, and that angle is normally reconstructed from *reference* currents rather than measured ones. Two questions remain open:

(i) How do we find the frame angle θ needed for the reference-frame transformation?

(ii) How do we determine a proper i^*_{ds} value, i.e. the one that maintains a desired constant flux level in the machine?

This document re-applies the same five derivation principles used for stator-flux oriented control, this time aligning the rotating frame with the *rotor* flux instead.

2 Why Rotor Flux?

Rotor-flux orientation is the classical choice for field-oriented control of induction machines. Its main strength is that the resulting torque and flux equations decouple *completely*: as will be shown, the leakage factor σ that complicates the stator-flux case disappears entirely, so i_{ds} and i_{qs} no longer cross-couple in either governing equation. This gives the best torque capability over the speed range from zero to nominal speed. The price is that the rotor time constant τ_r must be known reasonably accurately; and the scheme performs less well than stator-flux orientation once the machine is driven into field weakening.

3 Governing Machine Equations

Stator side

$$\overline{v}_s = r_s\overline{i}_s + p\,\overline{\lambda}_s + j\omega_e\overline{\lambda}_s, \qquad \overline{\lambda}_s = L_s\overline{i}_s + L_m\overline{i}_r \tag{1}$$

where $p = d/dt$, ω_e is the rotating speed of the chosen dq frame, and $L_s = l_{ls} + L_m$.

Rotor side

$$\overline{v}_r = r_r\overline{i}_r + p\,\overline{\lambda}_r + j(\omega_e - \omega_r)\overline{\lambda}_r = 0, \qquad \overline{\lambda}_r = L_r\overline{i}_r + L_m\overline{i}_s \tag{2}$$

where ω_r is the rotor shaft speed in electrical rad/s, $L_r = l_{lr} + L_m$, and $\overline{v}_r = 0$ because the rotor windings are short-circuited.

General torque relation

Independently of which flux is used for orientation, the torque can be written as

$$\tau = \frac{3}{2}\hat{p}\,\mathrm{Im}\Big(\overline{\lambda}_s^* \cdot \bar{i}_s\Big) \tag{3}$$

($\hat{p}$ here denotes the number of pole pairs, $\mathrm{Im}(\cdot)$ the imaginary part.) This is the same relation used in the stator-flux derivation; it will be re-expressed in terms of the rotor flux below.

4 Eliminating the Rotor Current (Principle 1)

Start from the two flux-linkage equations:

$$\overline{\lambda}_s = L_s\bar{i}_s + L_m\bar{i}_r, \qquad \overline{\lambda}_r = L_r\bar{i}_r + L_m\bar{i}_s \tag{4}$$

Solve the rotor equation for $\bar{i}_r$:

$$\bar{i}_r = \frac{\overline{\lambda}_r - L_m\bar{i}_s}{L_r} \tag{5}$$

This single relation will be reused twice below: once to rewrite the general torque expression in terms of rotor flux, and once to eliminate $\bar{i}_r$ from the rotor voltage equation.

Substituting (5) into the stator flux equation and introducing the leakage factor $\sigma = \dfrac{L_sL_r - L_m^2}{L_sL_r}$ (small, since $L_s = l_{ls} + L_m > L_m$ and $L_r = l_{lr} + L_m > L_m$; $\sigma \to 0$ as leakage $\to 0$) gives, exactly as in the stator-flux case,

$$\overline{\lambda}_s = \sigma L_s\bar{i}_s + \frac{L_m}{L_r}\overline{\lambda}_r \tag{6}$$

This is the same identity derived in the stator-flux walkthrough – it relates the two fluxes regardless of which one is chosen for orientation, and is reused here without re-deriving it.

5 Choosing the Torque Equation and the Orientation

Substitute (6) into the general torque relation (3):

$$\tau = \frac{3}{2}\hat{p}\,\mathrm{Im}\Big\{\big[\sigma L_s\bar{i}_s^* + \tfrac{L_m}{L_r}\overline{\lambda}_r^*\big]\cdot\bar{i}_s\Big\} = \frac{3}{2}\hat{p}\,\sigma L_s\,\mathrm{Im}\big(\bar{i}_s^*\bar{i}_s\big) + \frac{3}{2}\hat{p}\,\frac{L_m}{L_r}\,\mathrm{Im}\big(\overline{\lambda}_r^*\bar{i}_s\big) \tag{7}$$

The first term vanishes because $\bar{i}_s^*\bar{i}_s = |\bar{i}_s|^2$ is purely real, leaving

$$\tau = \frac{3}{2}\hat{p}\,\frac{L_m}{L_r}\,\mathrm{Im}\big(\overline{\lambda}_r^*\bar{i}_s\big) \tag{8}$$

Now align the d-axis of the rotating frame with $\overline{\lambda}_r$ itself. By construction of this frame,

$$\big|\overline{\lambda}_{dqr}\big| = \lambda_{dr}, \qquad \boxed{\lambda_{qr} \equiv 0} \tag{9}$$

which is the key simplifying condition used throughout the rest of the derivation. With $\overline{\lambda}_r^* = \lambda_{dr}$ (real) and $\bar{i}_s = i_{ds} + ji_{qs}$, the torque reduces to

$$T = \frac{3}{2}\hat{p}\,\frac{L_m}{L_r}\,\lambda_{dr}\,i_{qs} \tag{10}$$

the rotor-flux analogue of the PMSM-like torque equation found for stator-flux orientation – here λ_{dr} plays the role of the controllable flux, scaled by L_m/L_r.

Two tasks remain, mirroring Section 1:

① Find the frame angle $\theta_r = \angle\overline{\lambda}_r$, which (unlike the stator-flux case) is not obtained from a direct back-EMF integral, but from the measured/estimated rotor position together with an integrated slip angle (Section 9).

② Determine a proper i_{ds}^* that maintains a constant λ_{dr} → go back to the machine equations.

6 Eliminating the Rotor Current from the Rotor Voltage Equation (Principle 2)

The rotor voltage equation is

$$\overline{v}_r = r_r\overline{i}_r + (p + js\omega_e)\,\overline{\lambda}_r = 0 \tag{11}$$

(stator currents are always retained, since they are measurable and controllable). Substitute $\overline{i}_r$ from (5) directly – unlike the stator-flux case, no further reduction to $\overline{\lambda}_s$ is needed, since the target variable here is already $\overline{\lambda}_r$:

$$0 = \frac{r_r}{L_r}\left(\overline{\lambda}_r - L_m\overline{i}_s\right) + (p + js\omega_e)\overline{\lambda}_r \tag{12}$$

Defining the *rotor time constant*

$$\tau_r = \frac{L_r}{r_r}, \tag{13}$$

and multiplying through by τ_r, the equation collapses to one containing only the rotor flux, the stator current, and machine parameters:

$$\boxed{0 = \left(\overline{\lambda}_r - L_m\overline{i}_s\right) + \tau_r(p + js\omega_e)\overline{\lambda}_r} \tag{14}$$

This is the master equation of rotor-flux oriented control. Notice that no leakage factor σ appears anywhere – a first indication of the full decoupling property mentioned in Section 2.

7 Decomposition into d, q Components and the Orientation Simplification (Principle 3)

Writing $\overline{\lambda}_r = \lambda_{dr} + j\lambda_{qr}$ and combining coefficients, equation (14) becomes

$$0 = (1 + \tau_r p + js\omega_e\tau_r)\overline{\lambda}_r - L_m\overline{i}_s \tag{15}$$

which splits into real (d) and imaginary (q) parts:

$$0 = (1 + \tau_r p)\lambda_{dr} - s\omega_e\tau_r\lambda_{qr} - L_m i_{ds} \tag{16}$$
$$0 = (1 + \tau_r p)\lambda_{qr} + s\omega_e\tau_r\lambda_{dr} - L_m i_{qs} \tag{17}$$

Now apply the orientation condition from Section 5: $\lambda_{qr} = 0$. Every term containing λ_{qr} vanishes, leaving two compact equations that form the core of the controller:

$$0 = (1 + \tau_r p)\lambda_{dr} - L_m i_{ds} \tag{1}$$
$$0 = s\omega_e\tau_r\lambda_{dr} - L_m i_{qs} \tag{2}$$

Compare these with equations (1)–(2) of the stator-flux case: there, i_{ds} and i_{qs} were tangled together through σ-dependent cross terms. Here, (1) contains only i_{ds} and (2) contains only i_{qs} – the two channels are already decoupled, with no further algebra required.

8 Slip Estimation from Equation (2) (Principle 4)

Solving (2) for $s\omega_e$:

$$\boxed{s\omega_e = \frac{L_m\, i_{qs}}{\tau_r\, \lambda_{dr}}} \tag{3}$$

where i_{qs} comes from (or is set as) the reference current and λ_{dr} is obtained from (1) below (or assumed equal to its steady-state value $L_m i_{ds}$). This gives the slip frequency needed to correctly locate the rotating frame – the same role played by equation (3) in the stator-flux case, but now free of the leakage-dependent denominator.

9 The i_{ds} Reference from Equation (1) (Principle 5)

The reference i_{qs}^* is set directly by the torque command, since $T = \frac{3}{2}\hat{p}\,\frac{L_m}{L_r}\,\lambda_{dr}\, i_{qs}$. Equation (1) rearranges directly into the companion relation that gives the i_{ds} reference needed to hold λ_{dr} at its desired level:

$$\boxed{i_{ds} = \frac{1}{L_m}\big(1 + \tau_r p\big)\,\lambda_{dr}} \tag{4}$$

Unlike the stator-flux case, this relation contains no feed-forward term in i_{qs} at all – another consequence of the complete decoupling noted above. With a constant flux command λ_{dr}^* in steady state ($p = 0$), this reduces to the familiar $i_{ds}^* = \lambda_{dr}^*/L_m$.

10 Assembling the Control Structure

Equations (3) and (4) together define the controller:

- The flux reference λ_{dr}^* is passed through the lead filter $(1+\tau_r p)/L_m$ – equation (4) implemented directly – to produce i_{ds}^*. Because the relation is already decoupled, no cross-feed term from i_{qs} is required here, in contrast to the stator-flux controller.
- The torque command is scaled (via $\frac{2}{3\hat{p}} \cdot \frac{L_r}{L_m \lambda_{dr}}$) into i_{qs}^*.
- Together with i_{ds}^*, i_{qs}^*, and λ_{dr} (or its steady-state estimate $L_m i_{ds}$), the slip relation (3) is evaluated and integrated ($1/s$) to obtain the slip angle.
- The slip angle is added to the measured (or estimated) rotor electrical angle θ_{er} to obtain the total field angle used for the $dq \leftrightarrow abc$ frame transformation.

A flux observer with a PI correction loop, analogous to the $\Delta\lambda$ term used in the stator-flux controller, can still be added in practice to compensate for inaccuracies in L_m or τ_r. It is not structurally required by the equations themselves, however, since (1)–(2) contain no cross-coupling terms to absorb.

11 Comparison with Stator-Flux Orientation

The two derivations follow exactly the same five principles, but the outcomes differ in one essential respect: the leakage factor σ, which appears throughout the stator-flux equations, never appears in the rotor-flux equations. This is why rotor-flux orientation gives full decoupling between the torque- and flux-producing currents and the best torque capability from zero to nominal speed, at the cost of needing an accurate rotor time constant τ_r; stator-flux orientation instead trades some coupling complexity for better performance in field weakening and one fewer parameter dependency on L_m.

Summary of the Derivation

1. Write the general torque relation in terms of stator flux and stator current (valid for any orientation choice).
2. Eliminate the rotor current using the flux-linkage equations, and use the resulting identity to re-express the torque in terms of rotor flux and stator current (Principle 1).
3. Orient the rotating frame on $\overline{\lambda}_r$, forcing $\lambda_{qr} = 0$, and obtain the simplified torque equation.
4. Eliminate the rotor current from the rotor voltage equation directly, leaving an equation in $\overline{\lambda}_r$, $\overline{i}_s$ only (Principle 2).
5. Decompose into d, q and apply $\lambda_{qr} = 0$ to get two fully decoupled compact equations (Principle 3).
6. One equation gives the slip estimate $s\omega_e$, used to build the frame angle (Principle 4).
7. The other equation gives the i_{ds}^* reference needed to hold the flux constant (Principle 5).
8. Assemble both into the flux/torque control structure.